

# SEONGKU KANG

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## RESEARCH INTERESTS

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- Data mining and machine learning for real-world applications
- Information retrieval and recommender systems

## EDUCATION

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**Ph.D., Computer Science and Engineering, POSTECH** Mar 2018 - Aug 2023

- Advisor: [Prof. Hwanjo Yu](#)

**B.S., Computer Science and Engineering, NTU** Jul 2016 - May 2017

- Exchange Student

**B.S., Computer Science and Engineering, Hanyang University** Mar 2012 - Feb 2018

- Cumulative GPA: 4.43/4.50 (ranked 1st in CSE Dept.)

## SELECTED PUBLICATIONS

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[WWW'24] SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han  
Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy

[KDD'24] Gyuseok Lee\*, SeongKu Kang\*, Wonbin Kweon, Hwanjo Yu (co-first authors)  
Continual Collaborative Distillation for Recommender System

[CIKM'23-Short] Youngjune Lee, Keunchan Park, Yeongjong Jeong, SeongKu Kang (corresponding author)  
MvFS: Multi-view Feature Selection for Recommender System

[WWW'23] SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu  
Distillation from Heterogeneous Models for Top-K Recommendation

[WWW'22] SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu  
Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering

[KDD'21] SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
Topology Distillation for Recommender System

[CIKM'20] SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
DE-RRD: A Knowledge Distillation Framework for Recommender System

[CIKM'19] SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu  
Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users

Two first-authored papers under review at top-tier DM/NLP conferences

## EXPERIENCE

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**Data Mining Group, UIUC** Aug 2023 - Present

- Postdoctoral Researcher (advisor: [Prof. Jiawei Han](#))

**Microsoft Research Asia, Beijing, China** Oct 2022 - April 2023

- Research intern in Social Computing group (mentors: [Dr. Xing Xie](#), [Dr. Jianxun Lian](#))
- Award of excellence (Stars of Tomorrow, top 10%)

## AWARDS & FELLOWSHIPS

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### Awards

- Stars of Tomorrow Award for Outstanding Intern, Microsoft Research Asia, 2023
- POSTECH Research Award (4th place), National Research Foundation and POSTECH, Korea, 2022
- Music Playlist Recommendation Competition (3rd place), Kakao, 2020
- SIGIR Student Travel Award, ACM SIGIR, 2019-2020
- Dean's List, Hanyang University, 2018
- Best Paper Award (3rd place), Korean Database Conference (KDBC), 2017

### Fellowships

- POSTECHIAN Fellowship, POSTECH, 2023
- NAVER Ph.D Fellowship, NAVER, 2021
- NAVER Academic Scholarship, NAVER, 2017
- National Science and Technology Scholarship, Ministry of Education, Korea, 2012-2017
- Hanyang Brain Scholarship, Hanyang University, 2012

## TALKS

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|---------------------------------------------------------------------------------|----------|
| • CS/AI Seminar, Ulsan National Institute of Science and Technology (UNIST)     | Nov 2023 |
| • Invited Seminar, Electronic & Information Research Information Center (EIRIC) | Sep 2023 |
| • Invited Seminar, Social Computing Group, Microsoft Research Asia (MSRA)       | Jun 2023 |
| • Technology Showcase, Korea AI summit                                          | Dec 2022 |

## FULL PUBLICATIONS

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Unless otherwise specified, the paper is accepted as a long/regular paper. \* indicates equal contribution.

### Conferences

1. Continual Collaborative Distillation for Recommender System  
Gyuseok Lee\*, SeongKu Kang\*, Wonbin Kweon, Hwanjo Yu  
KDD'24 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
2. PEARL: A Review-driven Persona-Knowledge grounded Conversational Recommendation Dataset  
Minjin Kim\*, Minju Kim\*, Hana Kim, Beong-woo Kwak, Soyeon Jeon, Hyunseo Kim, SeongKu Kang,  
Youngjae Yu, Jinyoung Yeo, Dongha Lee  
ACL'24 Findings | Annual Meeting of the Association for Computational Linguistics
3. Self-Consistent Reasoning-based Aspect-Sentiment Quad Prediction with Extract-Then-Assign Strategy  
Jieyong Kim\*, Ryang Heo\*, Yongsik Seo, SeongKu Kang, Jinyoung Yeo, Dongha Lee  
ACL'24-short Findings | Annual Meeting of the Association for Computational Linguistics
4. Multi-Domain Sequential Recommendation via Domain Space Learning  
Junyoung Hwang, Hyunjun Ju, SeongKu Kang, Sanghwan Jang, Hwanjo Yu  
SIGIR'24 | ACM SIGIR Conference on Research and Development in Information Retrieval
5. Learning Discriminative Dynamics with Label Corruption for Noisy Label Detection  
Suyeon Kim, Dongha Lee, SeongKu Kang, Sukang Chae, Sanghwan Jang, Hwanjo Yu  
CVPR'24 | IEEE / CVF Computer Vision and Pattern Recognition
6. Improving Retrieval in Theme-specific Applications using a Corpus Topical Taxonomy  
SeongKu Kang, Shivam Agarwal, Bowen Jin, Dongha Lee, Hwanjo Yu, Jiawei Han  
WWW'24 | ACM The Web Conference

7. Top-Personalized-K Recommendation  
Wonbin Kweon, SeongKu Kang, Sanghwan Jang, Hwanjo Yu  
WWW'24 | ACM The Web Conference
8. Multi-Domain Recommendation to Attract Users via Domain Preference Modeling  
Hyunjun Ju, SeongKu Kang, Dongha Lee, Junyoung Hwang, Sanghwan Jang, Hwanjo Yu  
AAAI'24 | AAAI Conference on Artificial Intelligence
9. MvFS: Multi-view Feature Selection for Recommender System  
Youngjune Lee, Keunchan Park, Yeongjong Jeong, SeongKu Kang (corresponding author)  
CIKM'23-short | ACM International Conference on Information and Knowledge Management
10. Distillation from Heterogeneous Models for Top-K Recommendation  
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu  
WWW'23 | ACM The Web Conference
11. Learning Topology-Specific Experts for Molecular Property Prediction  
Suyeon Kim, Dongha Lee, SeongKu Kang, Seonghyeon Lee, Hwanjo Yu  
AAAI'23 | AAAI Conference on Artificial Intelligence
12. Consensus Learning from Heterogeneous Objectives for One-Class Collaborative Filtering  
SeongKu Kang, Dongha Lee, Wonbin Kweon, Junyoung Hwang, Hwanjo Yu  
WWW'22 | ACM The Web Conference
13. TaxoCom: Topic Taxonomy Completion with Hierarchical Discovery of Novel Topic Clusters  
Dongha Lee, Jiaming Shen, SeongKu Kang, Susik Yoon, Jiawei Han, Hwanjo Yu  
WWW'22 | ACM The Web Conference
14. Obtaining Calibrated Probabilities with Personalized Ranking Models  
Wonbin Kweon, SeongKu Kang, Hwanjo Yu  
AAAI'22 | AAAI Conference on Artificial Intelligence
15. Topology Distillation for Recommender System  
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
KDD'21 | ACM SIGKDD Conference on Knowledge Discovery and Data Mining
16. Bootstrapping User and Item Representations for One-Class Collaborative Filtering  
Dongha Lee, SeongKu Kang, Hyunjun Ju, Chanyoung Park, Hwanjo Yu  
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
17. Unsupervised Proxy Selection for Session-based Recommender Systems  
Junsu Cho, SeongKu Kang, Dongmin Hyun, Hwanjo Yu  
SIGIR'21 | ACM SIGIR Conference on Research and Development in Information Retrieval
18. Learning Heterogeneous Temporal Patterns of User Preference for Timely Recommendation  
Junsu Cho, Dongmin Hyun, SeongKu Kang, Hwanjo Yu  
WWW'21 | ACM International World-Wide Web Conference
19. Bidirectional Distillation for Top-K Recommender System  
Wonbin Kweon, SeongKu Kang, Hwanjo Yu  
WWW'21 | ACM International World-Wide Web Conference
20. DE-RRD: A Knowledge Distillation Framework for Recommender System  
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
CIKM'20 | ACM International Conference on Information and Knowledge Management

21. Deep Rating Elicitation for New Users in Collaborative Filtering  
Wonbin Kweon, SeongKu Kang, Junyoung Hwang, Hwanjo Yu  
WWW'20-short | ACM International World-Wide Web Conference
22. Multi-Modal Component Embedding for Fake News Detection  
SeongKu Kang, Junyoung Hwang, Hwanjo Yu  
IMCOM'20 | IEEE International Conf. Ubiquitous Information Management and Communication
23. Semi-Supervised Learning for Cross-Domain Recommendation to Cold-start Users  
SeongKu Kang, Junyoung Hwang, Dongha Lee, Hwanjo Yu  
CIKM'19 | ACM International Conference on Information and Knowledge Management
24. Densifying a Trust Network for Effective Collaborative Filtering  
SeongKu Kang, Jemin Wang, Yeon-Chang Lee, Sang-Wook Kim  
KDBC'17 | Korean DataBase Conference

## **Journals**

1. Unbiased, Effective, and Efficient Distillation from Heterogeneous Models for Recommender Systems  
SeongKu Kang, Wonbin Kweon, Dongha Lee, Jianxun Lian, Xing Xie, Hwanjo Yu  
ACM Transactions on Recommender Systems (2024.02)
2. Mitigating Viewpoint Sensitivity of Self-Supervised One-Class Classifiers  
Hyunjun Ju, Dongha Lee, SeongKu Kang, Hwanjo Yu  
Information Sciences (SCI) (2022.09)
3. Personalized Knowledge Distillation for Recommender System  
SeongKu Kang, Dongha Lee, Wonbin Kweon, Hwanjo Yu  
Knowledge-Based Systems (SCI) (2022.03)
4. Item-side Ranking Regularized Distillation for Recommender System  
SeongKu Kang, Junyoung Hwang, Wonbin Kweon, Hwanjo Yu  
Information Sciences (SCI) (2021.11)

## **PROFESSIONAL SERVICES**

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### **Program Committee/Reviewer**

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD): 2023, 2024, 2025
- ACM International World-Wide Web Conference (WWW): 2024, 2025
- AAAI Conference on Artificial Intelligence (AAAI): 2024, 2025
- SIAM Conference on Data Mining (SDM): 2024

### **Journal Reviewer**

- Transactions on Information Systems: 2024
- Transactions on Knowledge and Data Engineering: 2024
- Knowledge-Based Systems: 2023, 2024
- Neural Computing and Applications: 2022

## **PATENTS**

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### **Issued**

- Method and Apparatus for Improving the Accuracy of Speech Recognition Technology Based on Text Data Analysis for Deaf Students (Registration Number: 10-1988165-0000 | 2019.06.04 | South Korea)

### **Pending**

- Distillation from Heterogeneous Models for Top-K Recommendation (2023)
- Molecular Clustering Device And Method Based on Deep Learning Technology Using Molecular Topology Information (2023)

## REFERENCES

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Prof. Jiawei Han (PostDoc advisor), UIUC

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Prof. Hwanjo Yu (PhD advisor), POSTECH

hwanjoyupostech@gmail.com

Dr. Xing Xie (Internship mentor), MSRA

xingx@microsoft.com

Dr. Jianxun Lian (Internship mentor), MSRA

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